



IUS
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DES SCIENCES

Faculté des Sciences et des Technologies

FST

Laboratoire N° 1 du cours Administration des Systèmes Informatiques

Niveau : Licence III

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L'objectif de ce TD est de :

1. Installer un serveur Linux minimal
2. Configurer le réseau
3. Mettre en place l'accès distant sécurisé avec SSH
4. Effectuer la configuration de base d'un serveur

- **Installation d'Ubuntu Server**

Profile configuration [help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Votre nom : donsam

Your servers name: donsamsrvr
The name it uses when it talks to other computers.

Choisir un nom d'utilisateur : donsam

Choisir un mot de passe : ****

Confirmer votre mot de passe : ****

```
donsam@donsamsrvr: ~  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.56.109' (ED25519) to the list of known hosts.  
donsam@192.168.56.109's password:  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of dim. 08 mars 2026 02:09:28 UTC  
  
System load:          0.07  
Usage of /:           44.7% of 11.21GB  
Memory usage:         6%  
Swap usage:           0%  
Processes:            104  
Users logged in:      1  
IPv4 address for enp0s3: 10.0.2.15  
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fec9:4e16  
  
La maintenance de sécurité étendue pour Applications n'est pas activée.  
0 mise à jour peut être appliquée immédiatement.  
  
Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.  
Visitez https://ubuntu.com/esm ou exécutez : sudo pro status  
  
donsam@donsamsrvr:~$
```

- **Depuis votre machine physique, exécutez ces commandes via PowerShell**

```
donsam@donsamsrvr:~$ lsb_release -a  
No LSB modules are available.  
Distributor ID: Ubuntu  
Description:    Ubuntu 24.04.4 LTS  
Release:        24.04  
Codename:       noble  
donsam@donsamsrvr:~$
```

```
donsam@donsamserver:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c9:4e:16 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85766sec preferred_lft 85766sec
    inet6 fd17:625c:f037:2:a00:27ff:fec9:4e16/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86174sec preferred_lft 14174sec
    inet6 fe80::a00:27ff:fec9:4e16/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:bc:44:ec brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.109/24 metric 100 brd 192.168.56.255 scope global dynamic enp0s8
        valid_lft 566sec preferred_lft 566sec
    inet6 fe80::a00:27ff:febc:44ec/64 scope link
        valid_lft forever preferred_lft forever
donsam@donsamserver:~$
```

```
donsam@donsamserver:~$ ping -c 4 google.com
PING google.com (142.250.176.78) 56(84) bytes of data:
64 bytes from lcmiaa-al-in-f14.1e100.net (142.250.176.78): icmp_seq=1 ttl=255 time=52.1 ms
64 bytes from lcmiaa-al-in-f14.1e100.net (142.250.176.78): icmp_seq=2 ttl=255 time=53.8 ms
64 bytes from lcmiaa-al-in-f14.1e100.net (142.250.176.78): icmp_seq=3 ttl=255 time=52.4 ms
64 bytes from lcmiaa-al-in-f14.1e100.net (142.250.176.78): icmp_seq=4 ttl=255 time=96.4 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 52.138/63.705/96.445/18.913 ms
donsam@donsamserver:~$
```

```
donsam@donsamserver:~$ df -h
Filesystem                                Size  Used Avail Use% Mounted on
tmpfs                                     392M  1,1M  391M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv        12G   5,1G   5,7G  48% /
tmpfs                                     2,0G    0    2,0G   0% /dev/shm
tmpfs                                     5,0M    0    5,0M   0% /run/lock
/dev/sda2                                2,0G  104M   1,7G   6% /boot
tmpfs                                     392M   12K   392M   1% /run/user/1000
donsam@donsamserver:~$ |
```

```
donsam@donsamserver:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3,8Gi         429Mi         2,8Gi          1,1Mi          869Mi
Swap:          2,2Gi           0B          2,2Gi
donsam@donsamserver:~$ |
```

```
donsam@donsamserver:~$ sudo systemctl status ssh
[sudo] password for donsam:
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; pres>
   Active: active (running) since Sun 2026-03-08 02:08:07 UTC; 7min a>
  TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 2223 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0>
  Main PID: 2224 (sshd)
    Tasks: 1 (limit: 4601)
   Memory: 4.0M (peak: 5.0M)
      CPU: 77ms
   CGroup: /system.slice/ssh.service
           └─2224 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 sta>

mars 08 02:08:07 donsamservice systemd[1]: Starting ssh.service - OpenBS>
mars 08 02:08:07 donsamservice sshd[2224]: Server listening on 0.0.0.0 p>
mars 08 02:08:07 donsamservice sshd[2224]: Server listening on :: port 2>
mars 08 02:08:07 donsamservice systemd[1]: Started ssh.service - OpenBSD>
mars 08 02:09:20 donsamservice sshd[2228]: Accepted password for donsam >
mars 08 02:09:20 donsamservice sshd[2228]: pam_unix(sshd:session): sessi>
lines 1-20/20 (END)
```

```
donsam@donsamserver:~$ ps
  PID TTY          TIME CMD
 2284 pts/0    00:00:00 bash
 2329 pts/0    00:00:00 ps
```

```
donsam@donsamserver:~$ top
top - 02:16:33 up 14 min,  2 users,  load average: 0,00, 0,02, 0,04
Tasks: 104 total,  1 running, 101 sleeping,  2 stopped,  0 zombie
%Cpu(s):  0,0 us, 11,1 sy,  0,0 ni, 77,8 id,  0,0 wa,  0,0 hi, 11,1 si,
MiB Mem :  3916,3 total,  2845,4 free,   428,9 used,   870,1 buff/c
MiB Swap:  2276,0 total,  2276,0 free,    0,0 used.  3487,4 avail
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
128	root	20	0	0	0	0	I	9,1	0,0	0:05.92
2283	donsam	20	0	15128	7144	5144	S	9,1	0,2	0:02.34
1	root	20	0	22640	13772	9452	S	0,0	0,3	0:01.86
2	root	20	0	0	0	0	S	0,0	0,0	0:00.00
3	root	20	0	0	0	0	S	0,0	0,0	0:00.00
4	root	0	-20	0	0	0	I	0,0	0,0	0:00.00
5	root	0	-20	0	0	0	I	0,0	0,0	0:00.00
6	root	0	-20	0	0	0	I	0,0	0,0	0:00.00
7	root	0	-20	0	0	0	I	0,0	0,0	0:00.00
12	root	0	-20	0	0	0	I	0,0	0,0	0:00.00
13	root	20	0	0	0	0	I	0,0	0,0	0:00.00
14	root	20	0	0	0	0	I	0,0	0,0	0:00.00
15	root	20	0	0	0	0	I	0,0	0,0	0:00.00
16	root	20	0	0	0	0	S	0,0	0,0	0:00.48
17	root	20	0	0	0	0	I	0,0	0,0	0:00.10
18	root	rt	0	0	0	0	S	0,0	0,0	0:00.01
19	root	-51	0	0	0	0	S	0,0	0,0	0:00.00
20	root	20	0	0	0	0	S	0,0	0,0	0:00.00
21	root	20	0	0	0	0	S	0,0	0,0	0:00.00
22	root	0	-20	0	0	0	I	0,0	0,0	0:00.00
23	root	20	0	0	0	0	S	0,0	0,0	0:00.00
24	root	20	0	0	0	0	S	0,0	0,0	0:00.00
25	root	20	0	0	0	0	I	0,0	0,0	0:00.18
26	root	20	0	0	0	0	S	0,0	0,0	0:00.00
27	root	20	0	0	0	0	I	0,0	0,0	0:00.07

Conclusion

Ce TD vous a permis d'acquérir les compétences essentielles pour déployer et sécuriser un serveur Linux, depuis son installation minimale jusqu'à la configuration réseau et l'accès distant sécurisé via SSH.